

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Third Semester of B. Tech Examination
November 2019
IT216 / IT242 JAVA PROGRAMMING

Date: 06.11.2019, Wednesday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 Answer any Two: [08]

- a) Differentiate encapsulation and abstraction. Explain *abstract* keyword with example.
- b) What is lifetime of static, instance and local variables? Explain with examples.
- c) Explain the following *keywords*:
 a) *this* b) *super*

Q – 2.a Explain Array of Objects in java with given example. [03]

Create **Student** class with `int roll_no` and `String name`, create 5 student objects by using array of **Student** Class. Add method *printall()* to print all studnets information. Initialize all objects with proper values and print all informaiton.

Q – 2.b Explain String Class with an example of any two constructors. [04]

What will be the output of following code?

```
public class Test{
    public static void main (String[] args){
        String s1 = "CHARUSAT";
        String s2 = " Hello CHARUSAT ";
        System.out.println("Substring:" + s1.substring(5));
        System.out.println("Substring:" + s2.substring(5,8));
        System.out.println(s1.compareTo(s2.trim().substring(6, 14)));
    }
}
```

OR

Q – 2.b What will be the output of following codes? Justify your answer and correct the code if there is an Error or an Exception. [04]

```
a)
class Animal {
    private void eat() {
        System.out.println("Eating");
    }
}
class Cat extends Animal {
    public void eat() {
        System.out.println("Cat eats Fish");
    }
}
public class MainClass {
    public static void main(
        String args[]) {
        Animal c = new Cat();
        c.eat();
    }
}
```

```
b)
public class Sum {
    public int x, y;
    public double a, b;
    public void sum(Sum s) {
        sum(s.a, s.b);
    }
    public void sum(int x, int y) {
        System.out.println(x + y);
    }
    public void sum(double x, double y) {
        System.out.println((x + y));
    }
    public static void main(String args[]) {
        Sum s = new Sum();
        s.x = s.y = 10;
        s.a = s.b = 10.4;
        s.sum(s);
    }
}
```

Q – 2.c Write a Java program to demonstrate the functionality of Hybrid Inheritance with the help of interface. [06]

OR

Q – 2.c Differentiate final, finally and finalize. Explain each with small example. [06]

Q - 3 Answer any TWO. [14]

- a) Write a program to read single string of comma separated integer values as a command line argument and find out 2nd largest prime number among them, if no prime found print "Not Found" message.

e.g. Input: 12,45,78,11,35,22 Output: Not Found

Input: 10,17,13,16,11,2 Output: 13

- b) Design a class named **Fan** to represent a fan. The class contains:
- Three constants named SLOW, MEDIUM and FAST with values 1,2 and 3 to denote the fan speed.
 - An int data field named *speed* that specifies the speed of the fan (default SLOW).
 - A Boolean data field named *f_on* that specifies whether the fan is on (default *false*).
 - A double data field named *radius* that specifies the radius of the fan (default 4).
 - A data field named *color* that specifies the color of the fan (default *blue*).
 - A no-arg constructor that creates a default **Fan**.
 - A parameterized constructor initializes the **Fan** objects to given values.
 - A method named *display()* will display description for the fan. If the Fan is on, the *display()* method display should speed, color and radius. If the fan is not on, the method should return Fan color and radius along with the message "fan is off".

Write a test program that creates two **Fan** objects. One with default values and the other with medium speed, radius 6, color brown, and turned on status true. Display the descriptions for two created **Fan** objects.

- c) Write a java program according to given instruction to demonstrate the use of abstract class in Java. Create abstract class **Arithmetic** and provide three abstract methods in it, *add(int a, int b)*, *sub(int a, int b)* and *mul(int a, int b)*. Extend **Arithmetic** class in **MyCalculator** in which implement all abstract methods.

SECTION – II

Q - 4 Answer any Two: [08]

- What is Collection framework in Java? List benefits of collection framework.
- Explain following terms with example:
 - Anonymous Inner Class
 - Dynamic method dispatch
- Explain the life cycle of thread in java.

Q – 5.a What is a buffer? How does a buffer work with an output stream? How does this improve the performance of I/O operations? What does it mean to flush a buffer? [07]

OR

Q – 5.a What is a functional interface? What are the rules of defining a functional interface? What is a Lambda Expression, explain its syntax with example. [07]

Q – 5.b Explain generics in Java with an example. Create custom ArrayList class which can add and read elements from it. [06]

OR

Q – 5.b What are the differences between ArrayList and LinkedList? Which list should you use to insert and delete elements at the beginning of a list? **[06]**

Q – 6 **Attempt any Two:** **[14]**

- a) Write a Multithreading program for the fruit market. The farmer will be able to produce different types of fruits (Apple, Orange, Papaya, and Watermelon), and put them in the market to sell. The market has limited capacity and farmers have to stand in a queue if the capacity is exceeded to sell their fruits. Consumers can come to the market any time and purchase their desired fruits; and if the fruits they want to buy runs out, they are willing to wait until the supply of that kind is ready. (Hint: implementing this market will encounter the producer and consumer problem, and it probably needs multiple buffers for different kinds of fruits).

Output:

Producer has produced Apple

Consumer has consumed Apple

Producer has produced Orange

Consumer has consumed Orange

Producer has produced Papaya

Consumer has consumed Papaya

Producer has produced Watermelon

Consumer has consumed Watermelon

...

- b) Write a program that raises two exceptions. Specify two 'catch' clauses for the two exceptions. Each 'catch' block handles a different type of exception. For example, the exception could be 'ArithmeticException' and 'ArrayIndexOutOfBoundsException'. Display an informative message in the 'finally' block.

- c) Write a GUI application in java, which enter the details of a student and on the submit display the details of the student. (Student detail contains **Roll Number, Name, Gender, Department** etc.) Create two buttons in the applet like submit, and view. When the user clicks the submit button it stores the information of the student in the text file named "studentInfo.txt". When view button pressed it shows all the details of a students from "studentInfo.txt" to the text area.


